

Eunoia Junior College
9729 H2 Chemistry
2017 Paper 1 Suggested Solution

- 1 Let the percentage of ^{29}Si by $x\%$,

$$\frac{(92.23 \times 28) + 29x + 30(7.77 - x)}{100} = 28.10$$

$$233.1 - x = 227.56$$

$$x = 5.54$$

 $\Rightarrow \text{C}$
- 2 The most common isotopes are :
 ^7_3Li , ^9_4Be , $^{11}_5\text{B}$, $^{12}_6\text{C}$, $^{14}_7\text{N}$, $^{16}_8\text{O}$, $^{19}_9\text{F}$, $^{20}_{10}\text{Ne}$,
 $^{23}_{11}\text{Na}$ and $^{24}_{12}\text{Mg}$
1 ✓: ^7_3Li , ^9_4Be , $^{11}_5\text{B}$, $^{19}_9\text{F}$ and $^{23}_{11}\text{Na}$
2 ✓: $^{12}_6\text{C}$, $^{14}_7\text{N}$, $^{16}_8\text{O}$, $^{20}_{10}\text{Ne}$ and $^{24}_{12}\text{Mg}$
3 ✖: None of the above.
 $\Rightarrow \text{B}$
- 3 $\text{H}^{\sigma}_{\sigma}\text{C}^{\sigma/\pi} \Rightarrow \text{sp}^2$ hybridised $\Rightarrow$ trigonal planar
 $\text{H} \Rightarrow 120^\circ$
 $\Rightarrow \text{D}$
- 4 **1 ✓:** Boiling point of simple molecular compounds is a reflection of the strength of *intermolecular forces*, where the hydrogen bonds between H_2O molecules are stronger than the dispersion forces between CH_4 .
2 ✖: Boiling point of simple molecular compounds is a reflection of the strength of intermolecular forces and **not** the strength of covalent bonds.
3 ✖: H_2O and CH_4 both contains 10 electrons each
 $\Rightarrow \text{C}$
- 5 **A ✖:** Increasing the pressure does not causes the temperature to change.
B ✖: K.E. of a gas is dependent on the temperature, T , only, i.e. $\text{K.E.} \propto T$. Since T does not change, K.E. also does not change with pressure.
C ✖: The size of molecules is not affected by changes in the pressure.
D ✓: Increasing the pressure forces the molecules closer together, thereby increasing the strength of the intermolecular forces of attraction which are electrostatic in nature.
 $\Rightarrow \text{D}$
- 6 **A ✖:** H^+ is an electron-pair acceptor (Lewis acid).
B ✓: H_2O can donate its lone pair (Lewis base) and can dissociate to produce H^+ (Brønsted-Lowry acid).
C ✖: O^{2-} is an electron-pair donor (Lewis base) and a H^+ acceptor (Brønsted-Lowry base).
D ✖: OH^- is an electron-pair donor (Lewis base) and a H^+ acceptor (Brønsted-Lowry base).
 $\Rightarrow \text{B}$
- 7 Due to the increase in ENC (nuclear charge increases while shielding effect remains relatively constant), atomic radius decreases while electronegativity increases across the period

- Atomic radius (in nm):
 $\text{P}(0.110) < \text{Si}(0.117) < \text{Al}(0.143) < \text{Mg}(0.160)$
 Electronegativity: $\text{P} > \text{Si} > \text{Al} > \text{Mg}$
 $\Rightarrow \text{B}$
- 8 There is a sharp drop in the fifth IE from R to S
 $\Rightarrow$ the electron removed from S^{4+} belongs to a higher energy principal quantum shell.
 R^{4+} thus have a noble gas $ns^2 np^6$ electronic configuration and Q^{4+} have an electronic configuration of $ns^2 np^5$.
 Hence, Q prefer to form the Q^{3+} ion with a noble gas $ns^2 np^6$ electronic configuration.
 $\Rightarrow \text{B}$
- 9 Polarising power $\propto$ Charge density $\propto \frac{q^+}{r_+}$
 Mg^{2+} has the highest charge and smallest size among the 4 cations (Ba^{2+} , Cs^+ , Mg^{2+} and Na^+), hence the most polarising.
 $\Rightarrow \text{B}$
- 10 **1 ✓:** The process is thermodynamically feasible since $\Delta G^\ominus$ is negative. The process does not takes place at room temperature and pressure due to **very high activation energy, E_a** , resulting in an extremely small rate constant $\left(k = Ae^{-\frac{E_a}{RT}} \right)$.
2 ✖: $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$
 $= -1900 - 293(+3.4)$
 $= -2.9 \text{ kJ mol}^{-1}$
 However, a negative $\Delta G^\ominus$ implies a thermodynamically **feasible** process, which should take place on its own.
3 ✖: Since $\Delta G^\ominus = -2.9 \text{ kJ mol}^{-1}$, the **forward** reaction is spontaneous.
 $\Rightarrow \text{B}$
- 11 Given the first-order reaction

$$\text{X}_2(\text{g}) \xrightarrow{t_{1/2}=30 \text{ min}} 2\text{X}(\text{g})$$
1 ✓: 25% of X_2 would have remained after 2 half-lives ($= 2 \times 30 = 60 \text{ min}$):

$$\text{X}_2(\text{g}) \xrightarrow{30 \text{ min}} \frac{1}{2}\text{X}_2(\text{g}) \xrightarrow{30 \text{ min}} \frac{1}{4}\text{X}_2(\text{g})$$
 Meaning that 75% of X_2 had reacted
2 ✓: When 75% of X_2 has been converted, means 0.75 mol of X_2 reacted, so amt of X formed $= 2 \times 0.75 = 1.5 \text{ mol}$
3 ✓: Total amt of gaseous particles = amt of X_2 + amt of X $= 0.25 \text{ mol} + 1.5 \text{ mol} = 1.75 \text{ mol}$.
 Using the gas law at constant V and T , we have $\frac{p_1}{n_1} = \frac{p_2}{n_2}$. So

$$\text{pressure} = \frac{p}{1} \times 1.75 = 1.75p = \frac{7p}{4}$$
 $\Rightarrow \text{A}$
- 12 The rate equation follows the **molecularity** of the **slow step**, but only involving the **reactants**. Concentration term involving any intermediates must be replaced.
 $\Rightarrow \text{A}$

- 13
$$[\text{H}_2\text{O}] = \frac{\text{density}}{\text{molar mass}} = \frac{0.997 \text{ g cm}^{-3}}{18 \text{ g mol}^{-1}}$$

$$= \frac{0.997}{18} \text{ mol cm}^{-3}$$

$$= \frac{997}{18} \text{ mol dm}^{-3}$$
 In pure water, $[\text{H}_3\text{O}^+] = [\text{OH}^-]$. So

$$K_c = \frac{[\text{H}_3\text{O}^+]^2}{[\text{H}_2\text{O}]}$$

$$[\text{H}_3\text{O}^+] = \sqrt{K_c} [\text{H}_2\text{O}]$$

$$\eta_{\text{H}_3\text{O}^+} = [\text{H}_3\text{O}^+] \times 1.00 \text{ dm}^3$$

$$= \sqrt{K_c} [\text{H}_2\text{O}] \text{ mol}$$
 No. of H_3O^+ ions $= \eta_{\text{H}_3\text{O}^+} \times L$

$$= L\sqrt{K_c} [\text{H}_2\text{O}]$$

$$= \frac{997}{18} L\sqrt{K_c}$$
 $\Rightarrow \text{D}$
- 14 Using $\Delta G^\ominus = -RT \ln K$, $K = \frac{[\text{H}_2][\text{CO}_2]}{[\text{H}_2\text{O}][\text{CO}]}$.
 For $[\text{H}_2] = [\text{CO}_2] > [\text{H}_2\text{O}] = [\text{CO}]$, $K > 1$, so
 $\ln K > 0$
 $-RT \ln K < 0$
 $\Delta G^\ominus < 0$
 At 2, $\Delta G^\ominus = 0$, $[\text{H}_2] = [\text{CO}_2] = [\text{H}_2\text{O}] = [\text{CO}]$
 $\Rightarrow \text{D}$
- 15 $\text{PbCrO}_4(\text{s}) \rightleftharpoons \text{Pb}^{2+}(\text{aq}) + \text{CrO}_4^{2-}(\text{aq})$

$$[\text{Pb}^{2+}(\text{aq})] = [\text{CrO}_4^{2-}(\text{aq})]$$

$$= 1.3 \times 10^{-7} \text{ mol dm}^{-3}$$

$$K_{\text{sp}}(\text{PbCrO}_4) = [\text{Pb}^{2+}(\text{aq})] \times [\text{CrO}_4^{2-}(\text{aq})]$$

$$= (1.3 \times 10^{-7} \text{ mol dm}^{-3})^2$$

$$= 1.69 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$$
 $\Rightarrow \text{A}$
- 16 **1 ✓:**

$\begin{array}{c} \text{NH}_2 \\ \\ \text{CH}_3 - \text{CH} - \text{CH}_3 \end{array}$ 2-aminopropane $\text{C}_3\text{H}_9\text{N}$	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ \\ \text{Br} \end{array}$ 2-bromo-2-methylpropane $\text{C}_4\text{H}_9\text{Br}$
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2 ✓:

$\text{C}_2\text{H}_5 - \text{O} - \overset{\text{O}}{\parallel} - \text{C}_2\text{H}_5$ ethylpropanoate $\text{C}_5\text{H}_{10}\text{O}_2$	$\text{C}_2\text{H}_5 - \overset{\text{OH}}{\underset{ }{\text{CH}}} - \overset{\text{OH}}{\underset{ }{\text{CH}_2}}$ butane-1,2-diol $\text{C}_4\text{H}_{10}\text{O}_2$
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3 ✖:

$\text{C}_3\text{H}_7 - \text{C} \equiv \text{N}$ butanenitrile $\text{C}_4\text{H}_7\text{N}$	$\begin{array}{c} \text{CH}_3 \quad \text{O} \\ \quad \parallel \\ \text{CH}_3 - \text{CH} - \text{C} - \text{H} \end{array}$ 2-methylpropanal $\text{C}_4\text{H}_8\text{O}$
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 $\Rightarrow \text{B}$
- 17 H_2O is oxidised to O_2 at the anode:

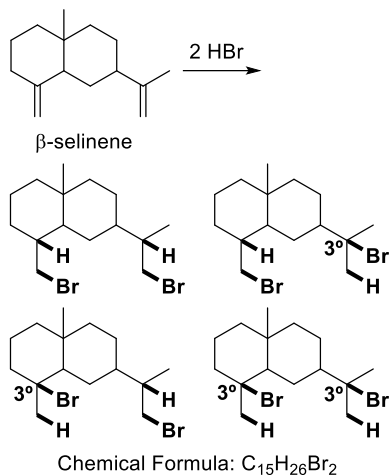
$$2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$$
 Overall equation is

$$4\text{CH}_2=\text{CHCN} + 2\text{H}_2\text{O} \rightarrow 2\text{NC}(\text{CH}_2)_4\text{CN} + \text{O}_2$$

$$\eta_{\text{O}_2} = \frac{1}{4} \eta_{\text{acrylonitrile}} = \frac{1}{4} \times 0.01 = 0.0025 \text{ mol}$$

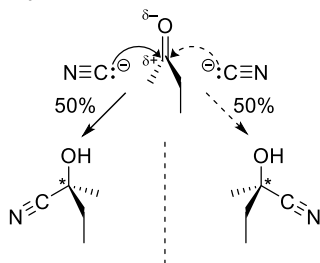
$$V_{\text{O}_2} = \eta_{\text{O}_2} \times V_m = 0.0025 \times 24000 = 60 \text{ cm}^3$$
 $\Rightarrow \text{A}$

- 18 β -Selinene, $C_{15}H_{24}$, reacts with 2 moles of HBr to give 4 addition products:



$\Rightarrow$ D

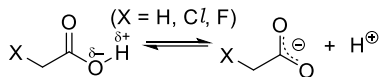
- 19 Only the reaction of the trigonal planar carbonyl group with HCN will give a mixture of the two enantiomers, as there is **equal probability** of the CN^- approaching the carbonyl C from either side:



The α -carbon in chloroalkane **B**, **C** and **D** are achiral to begin with and substitution of the Cl by OH will not give a chiral product.

$\Rightarrow$ A

- 20 The extent of dissociation :



depends on

- i) strength of the O-H bond

The weaker the O-H bond, the stronger the acid, the smaller pK_a is.

- ii) stability of the resultant carboxylate ion

The more stable the carboxylation ion, the stronger the acid, the smaller pK_a is.

- 1 $\checkmark$: F and Cl are more electronegative than H, drawing electrons away from the O-H bond, weakening the bond, facilitating dissociation.

- 2 $\times$: Being electronegative, both F and Cl helps to disperse the negative charge by drawing electrons towards themselves, **stabilising** the anion.

- 3 $\times$: Electron-donating inductive effect by the methyl group intensifies the negative charge, **destabilising** the carboxylate anion.

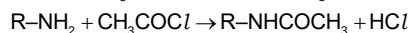
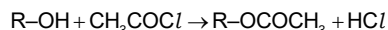
$\Rightarrow$ C

- 21 Order of Lewis basicity :

Amide < phenylamine < NH_3 < 1° amine < 2° amine < 3° amine

$\Rightarrow$ A

- 22 Ethanoyl chloride, CH_3COCl , reacts with the -OH (alcohols and phenols) and -NH- (amines and phenylamines) groups to give esters and amides respectively.



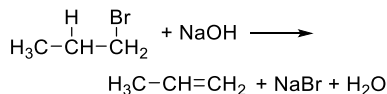
$\Rightarrow$ A

- 23 Halogenoarenes do not under nucleophilic substitution, as the C-X bond is strengthened due to delocalisation of the lone pair on X into the benzene ring, hence W and Y will have no reaction.

Among halogenalkanes, rate of reaction depends on the strength of the C-X bond; the weaker the bond, the faster the reaction. Hence the C-I bond in Z being weaker than the C-Cl bond in X will form yellow AgI first.

$\Rightarrow$ D

- 24 1 $\checkmark$: Elimination is usually a side reaction in nucleophilic substitution:

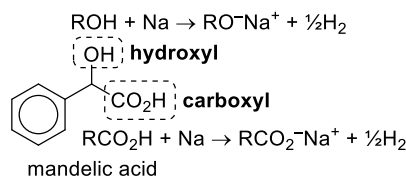


- 2 $\times$: Since there is only one Br in the starting material, there can only be one OH in the product.

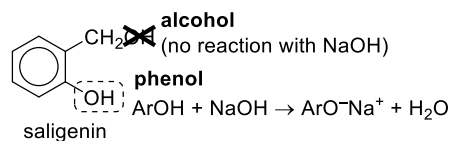
- 3 $\times$: The Br is on C-1 of the propane chain, hence substitution will give propan-1-ol and not propan-2-ol.

$\Rightarrow$ C

- 25 Both the hydroxyl and carboxyl groups in mandelic acid will react with Na(s), hence 2 moles of Na(s) is needed:

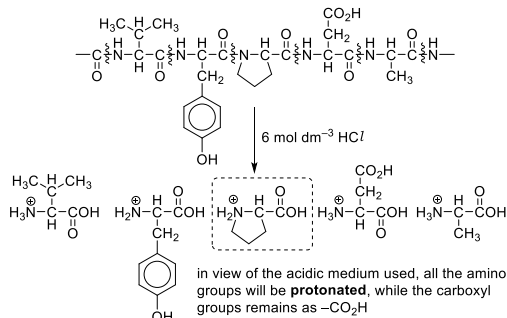


Only the phenolic -OH is acidic to react with NaOH(aq), but not the primary alcohol -CH2OH, hence 1 mole of NaOH(aq) is needed:



$\Rightarrow$ D

- 26 Hydrolysis of the peptide gives:

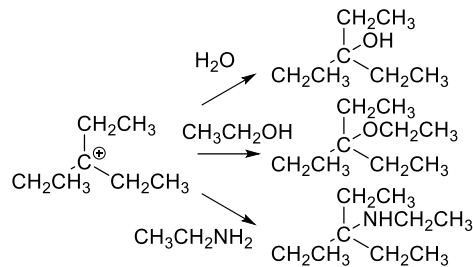


$\Rightarrow$ A

- 27 Solvents which are polar must contain polar bonds/groups, e.g. -OH and -NH2 groups. Only C_6H_{14} is non-polar.

As the carbocation intermediate is highly electrophilic, it will readily react with nucleophiles to give an adduct.

H_2O , CH_3CH_2OH and $CH_3CH_2NH_2$ all possesses lone pairs of electrons and can react with the carbocation:



However, the product from the reaction of H_2O is the **same** as that formed from the reaction of OH^- .

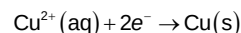
$\Rightarrow$ A

- 28 There is no water in pure ethanoic acid and liquid HCl, and hence there are very little ions as both compounds undergo very little or no (self-)dissociation.

Ethanoic acid is a weak acid which only partially dissociates, while HCl is a strong acid which completely dissociates, in **water** to give hydrated ions. Hence, 1 mol dm^{-3} HCl gives the most number of mobile ions and will be the best conductor of electricity.

$\Rightarrow$ B

- 29 The amount of copper deposited on the cathode depends on the amount of electrons transferred:



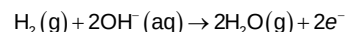
The amount of electrons transferred is dependent on the charge transferred, Q , which is given by

$$\text{charge}(Q) = \text{current}(I) \times \text{time}(t)$$

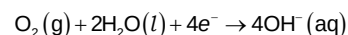
Hence, the number of atoms of Cu deposited will be determined by the current and time.

$\Rightarrow$ C

- 30 [Anode] $H_2(g)$ is **oxidised** to $H_2O(g)$:



[Cathode] $O_2(g)$ is **reduced** to $OH^-(aq)$:



$$E^\ominus(O_2|OH^-) = +0.40 V$$

Hence electrons will flow from the anode X to the cathode Y in the external circuit.

$\Rightarrow$ A

Answer Key

Qn	Ans
1	C
2	B
3	D
4	C
5	D
6	B
7	B
8	B
9	C
10	B

Qn	Ans
11	A
12	A
13	D
14	D
15	A
16	B
17	A
18	D
19	A
20	C

Qn	Ans
21	A
22	A
23	D
24	C
25	D
26	A
27	A
28	B
29	C
30	A